Research Mini-Project Report

Title: Integration of GitHub, Mendeley, and Google Colab for Data Analysis.

Aim: To integrate **GitHub**, **Google Colab**, and **Mendeley Reference Manager** in a research mini-project by performing data analysis, managing project files, and handling references.

Theory: Modern research projects require efficient tools for coding, collaboration, and referencing. **GitHub** is used for version control and storing project files.
Google Colab provides a cloud-based environment to write and execute Python code.
Mendeley Reference Manager helps in organizing research papers and generating citations automatically. The integration of these tools improves workflow, collaboration, and accuracy in research.

Dataset Description: The dataset used is the California Housing dataset, which contains information about housing prices based on various features such as median income, housing age, population, and location.
It is useful for analyzing relationships between income and house values.

Procedure:

Step 1: Data Analysis using Google Colab

- Open **Google Colab**
- Load the dataset
- Perform analysis using Python

Step 2: Code Implementation

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv('/content/sample_data/california_housing_test.csv')

# Display data
print("First 5 rows:")
print(df.head())

# Additional insights
print("\nAverage House Value:", df["median_house_value"].mean())
print("Maximum House Value:", df["median_house_value"].max())
print("Minimum House Value:", df["median_house_value"].min())
```

```
First 5 rows:
   longitude  latitude  housing_median_age  total_rooms  total_bedrooms  \
0    -122.05     37.37                27.0       3885.0           661.0
1    -118.30     34.26                43.0       1510.0           310.0
2    -117.81     33.78                27.0       3589.0           507.0
3    -118.36     33.82                28.0         67.0            15.0
4    -119.67     36.33                19.0       1241.0           244.0

   population  households  median_income  median_house_value
0      1537.0       606.0         6.6085            344700.0
1       809.0       277.0         3.5990            176500.0
2      1484.0       495.0         5.7934            270500.0
3        49.0        11.0         6.1359            330000.0
4       850.0       237.0         2.9375             81700.0

Average House Value: 205846.275
Maximum House Value: 500001.0
Minimum House Value: 22500.0
```

```python
# 1. Bar Graph (Top 10 house prices)
top10 = df.sort_values("median_house_value", ascending=False).head(10)
plt.figure()
plt.bar(range(len(top10)), top10["median_house_value"])
plt.title("Top 10 House Prices")
plt.xlabel("Index")
plt.ylabel("House Value")
plt.show()

# 2. Line Graph (Income vs House Value)
df_sorted = df.sort_values("median_income")
plt.figure()
plt.plot(df_sorted["median_income"], df_sorted["median_house_value"])
plt.title("Income vs House Value")
plt.xlabel("Median Income")
plt.ylabel("House Value")
plt.show()

# 3. Pie Chart (House Price Categories)
df["price_category"] = pd.cut(
    df["median_house_value"],
    bins=[0,100000,300000,500000],
    labels=["Low", "Medium", "High"]
)
plt.figure()
df["price_category"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("House Price Distribution")
plt.ylabel("")
plt.show()
```

Step 3: Upload to GitHub

- Create repository in **GitHub**
- Upload dataset, notebook, and report

Step 4: Reference Management

- Import research papers in **Mendeley Reference Manager**
- Organize references and edit metadata

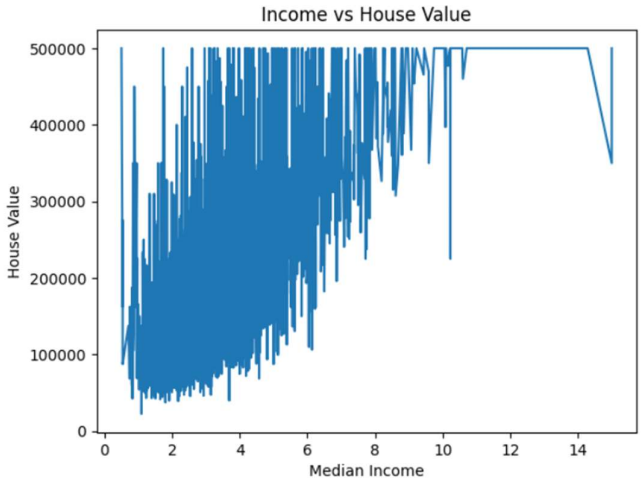
Step 5: Citation and Bibliography

- Use Mendeley plugin in **Microsoft Word**
- Insert citations and generate bibliography

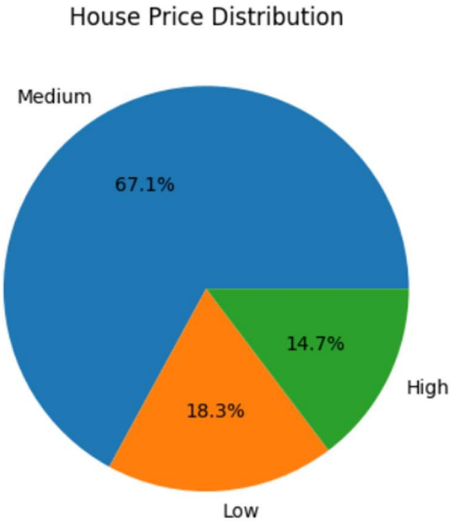
Output:

- Dataset successfully loaded and analyzed
- Bar graph showing top house prices

- Line graph showing relationship between income and house value



- Pie chart showing distribution of house price categories



Observation: The analysis shows that higher median income is associated with higher house prices. Most houses fall in the medium price category, while fewer belong to the high-price category.

Conclusion: Thus, the integration of **GitHub**, **Google Colab**, and **Mendeley Reference Manager** was successfully achieved. These tools helped in efficient data analysis, version control, and reference management.

References:

(Generated using Mendeley)

[1] A. Author, “Housing Price Analysis Using Machine Learning,” 2022.

[2] B. Author, “Data Analysis with Python,” Journal of Data Science, 2021.